Astronomical Image Denoising Using Physics-Guided Multi U-Net Models

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***Abstract*—Multiple noise sources originating from photon arrival statistics, detector electronics, and thermally generated charge accumulation inherently degrade astronomical images acquired by space-based telescopes. These noise processes often disproportionately obscure faint astrophysical structures and fine spatial details, limiting the scientific utility of the observational data and hence rendering robust denoising a critical preprocessing task. The proposed framework presents a physics-guided astronomical image denoising method combining domain-aware noise modeling with a multi-expert deep learning architecture. The term physics-guided denotes the use of physically grounded noise models during training data generation, rather than explicit physical constraints in the network architecture or loss. The framework employed multiple U-Net-based autoencoder models, each of which was trained to specialize in the suppression of a particular noise component-namely photon shot noise, readout noise, dark-current noise, and their composite interaction-using synthetically corrupted data generated via a physics-guided noise simulator. The simulator explicitly incorporates Poisson-distributed photon statistics and Gaussian electronic noise processes that enable physically realistic and astrophysically consistent synthesis of noise during training. In order to leverage the complementary strengths of the noise-specific denoisers, a lightweight learnable 1×1 convolutional fusion module fuses their outputs into a unified reconstruction. The experiments evaluate the proposed approach on curated Hubble Space Telescope imagery; it consistently outperforms single denoisers by yielding peak signal-to-noise ratios (PSNR) of 32.90 dB and a structural similarity index (SSIM) of 0.840. These results demonstrate the effectiveness of combining physics-guided noise simulation with multi-model deep learning and learned fusion for reconstructing high-fidelity images, preserving faint celestial structures across diverse astrophysical noise conditions.**

***Keywords*—Astronomical imaging, image denoising, U-Net, autoencoders, physics-based noise modeling, Hubble Space Telescope, convolutional neural networks, PSNR, SSIM.**

I. Introduction

Astronomical imaging makes distant galaxies, nebulae, and other structures observable; therefore, it is one of the cornerstones of observation-based astrophysics. Nonetheless, images taken by space-based telescopes are inherently degraded due to several types of noise caused by sensor physics and operational limitations. Photon shot noise has its origin in the stochastic nature of photon arrivals, read-out noise results from electronic amplification and digitization of charge carriers, whereas dark-current noise is caused by thermally generated electrons that accumulate in the detector [1]. These noise processes obscure faint astrophysical and low-intensity features disproportionately. Therefore, effective denoising is a key preprocessing step in astronomical data processing pipelines [4].

Various classical denoising techniques, such as anisotropic diffusion [5] and sparse coding–based dictionary learning [6], have been moderately successful on astronomical imagery. Although computationally efficient, these methods are based on handcrafted priors or local smoothing assumptions and, as a result, often yield an oversuppression of fine spatial details. In addition, they cannot robustly model the signal-dependent Poisson–Gaussian noise characteristics generated by real Charge-Coupled Device (CCD) sensors, which limits their performance when operating under photon-limited or mixed-noise conditions.

Recent deep learning advances have significantly improved performance in image restoration tasks, and CNNs now outperform classical approaches in image denoising, low-light enhancement, and inverse imaging problems [8]. Among those, U-Net architectures have achieved remarkable reconstruction performance thanks to their end-to-end encoder–decoder structure and skip connections, which model both multiscale features and spatial context [12]. Despite the success of these approaches, many learning-based denoising methods rely on simplified assumptions about Gaussian noise and neglect the underlying photon statistics and sensor physics inherent to the problem of astronomical image formation [3]. This mismatch can result in suboptimal generalization when applied to real observational data.

These limitations encourage the formulation of physics-guided denoising frameworks that integrate domain awareness during model design and training. In this context, a physics-guided astronomical image denoising framework is introduced in which four U-Net based autoencoders are trained separately. Three are specialized for photon shot noise, read-out noise, and dark-current noise and one trained on composite noise, using synthetically corrupted data generated through realistic CCD noise models [1]. A lightweight learnable 1×1 convolutional fusion module combines the outputs of these noise-specific experts into a unified denoised reconstruction, leveraging their complementary strengths inspired by the ensemble-based restoration strategies. [16].

Experiments are conducted using curated Hubble Space Telescope imagery retrieved from publicly available repositories [17]. The quality of restoration is quantitatively measured in terms of PSNR and SSIM, two widely adopted metrics for the quantitative evaluation of reconstruction fidelity and preservation of structural information in photon-constrained imaging problems in astronomy.

The main contributions of the proposed approach are summarized as follows.

- The developed denoising approach includes a physics-guided noise simulation strategy; it models Poisson-distributed photon shot noise along with Gaussian read-out and dark-current noise, thus enabling physically realistic training data generation.
- It presents a multi-expert denoising architecture which features multiple U-Net–based models independently specialized for different CCD noise components, together with a composite-noise baseline network.
- A lightweight learnable 1×1 convolutional fusion module which fuses the outputs of several learned noise-specific denoisers to effectively exploit their complementary reconstruction properties.
- Quantitative evaluation on curated Hubble Space Telescope imagery demonstrates that this fusion-based approach consistently outperforms the respective denoising models with respect to PSNR and SSIM while preserving faint astrophysical structures.

The rest of this paper is organized as follows-

Section II surveys previous work on classical, learning-based, and physics-guided astronomical image denoising approaches. Section III introduces the proposed physics-guided denoising framework, detailing the noise modeling process, U-Net architecture, training strategy. and learnable fusion mechanism. Section IV describes the experimental setting and presents both Quantitative and qualitative assessment of the proposed approach. Finally, Section V concludes the paper by outlining potential directions for future research.

II. Related Work

Early methods for image denoising have mainly applied various classical signal processing methods, including Gaussian smoothing, median filtering, and Wiener filtering. While being computationally efficient, these spatial filters tend to blur fine astrophysical structures and perform poorly in situations involving signal-dependent noise distributions or compound noise interactions, as often encountered within typical astronomical imaging scenarios. Their dependence on local averaging or linear smoothing inherently suppresses high-frequency information. These methods are thus not well-suited for preserving delicate celestial features embedded within noisy observations.

Further work developed physics-based formulations to better model the characteristics of observational data. The imaging sensors in telescopes have several noise components resulting from Poisson-distributed photon arrival fluctuations, Gaussian read-out noise during charge digitization, and thermally induced dark-current noise, which is integrated over exposure durations [1]. Such physically grounded models have been utilised to generate high-noise training data under realistic low-exposure conditions in order to improve the generalization capability of learning-based restoration algorithms [3].

With the advent of deep learning, architectures like DnCNN [8], FFDNet [10], and IRCNN [11] have significantly improved the performance of image denoising systems by outperforming classical approaches due to much deeper feature representations combined with end-to-end learning strategies. Autoencoder-based methods have further improved the reconstruction fidelity by exploiting latent-space features and nonlinear mappings optimized for the statistics of a particular noise type [6]. Of these variants, it is observed that U-Net-based architectures are particularly suited for astronomical denoising problems due to their multiscale feature aggregation along with skip connections that help retain spatial information even from severely degraded input images [12].

Recent works have targeted astronomical image reconstruction specifically using deep neural networks. U-Net–based architectures have been used for suppressing the CCD sensor noise, restoring faint galaxy structures, and improving low-light astrophysical observations [7], [13]. A notable example is the multi-model denoising system by Elhakiem et al., which trains U-Net models for dark-current, read-out, shot, and composite noise components in an independent fashion and merges their outputs through a learned integration mechanism [14]. Other studies have adopted physics-aware noise simulation, hybrid learning strategies, and domain-specific architectural modifications to improve the fidelity of astronomical reconstructions under diverse noise conditions [16].

III. Methodology

The proposed denoising framework combines physics-grounded noise modeling with multiple U-Net–based autoencoders and a learnable fusion mechanism to reconstruct clean astronomical images from noise-degraded observations. The dataset, noise simulation process, network design, training strategy, and inference pipeline are explained in this section to constitute the overall methodology.

A. Dataset

The proposed framework uses the Top 100 Hubble Telescope Images dataset available on Kaggle and provided by Redwan Karim Sony [17]. The dataset includes 100 high-quality images of galaxies, nebulae, star clusters, and other deep-sky objects captured by the Hubble Space Telescope. To maintain consistency, all images were converted to grayscale and then resized preserving aspect ratios constraining the longer edge to 512×512 pixels.

A patch-based sampling strategy was employed to extend the effective training corpus. Every image was decomposed into several randomly selected 128×128 patches, thereby generating approximately 3900 training patches per epoch over the full dataset. Noisy input patches are generated dynamically with the use of a physics-guided noise simulator, allowing the construction of realistic noisy–clean pairs emulating CCD-based astronomical imaging conditions.

B. Physics-Based Noise Models

The astronomical images captured by CCD sensors are indeed affected by several kinds of noise, arising from photon statistics and sensor electronics. To more precisely mimic real observational degradation patterns, a physics-based simulator was implemented that introduces astrophysically realistic noise into clean Hubble patches. The simulator models three main noise components.

1) Photon Shot Noise: Photon shot noise originates from the quantized nature of photon arrivals and is modeled as a Poisson counting process. Given a clean image I and exposure time t, and a photon conversion factor α, the expected photon count is

$$\lambda = I \cdot t \cdot \alpha \tag{1}$$

Photon shot noise is introduced by Poisson sampling of expected photon counts,

$$N_{\text{shot}} \sim \text{Poisson}(\lambda) \tag{2}$$

which replaces the clean signal in the electron domain rather than being added additively.

2) Read-Out Noise: Read-out noise results from electronic charge amplification and analog-to-digital conversion processes within CCD circuitry. It is modeled as an additive Gaussian process independent of signal intensity:

$$N_{\text{ron}} \sim \mathcal{N}(0, \sigma_{\text{ron}}^2) \tag{3}$$

where σ_{ron} denotes the read-out standard deviation.

3) Dark-Current Noise: Dark-current noise arises from thermally generated electrons accumulating in each pixel during exposure. A Gaussian approximation is adopted as a commonly accepted assumption in regimes where thermally generated electron counts are sufficiently high. It is modeled as

$$N_{\text{dc}} \sim \mathcal{N}(0, \sigma_{\text{dc}}^2 \cdot t) \tag{4}$$

where σ_{dc} represents the sensor's dark-current coefficient.

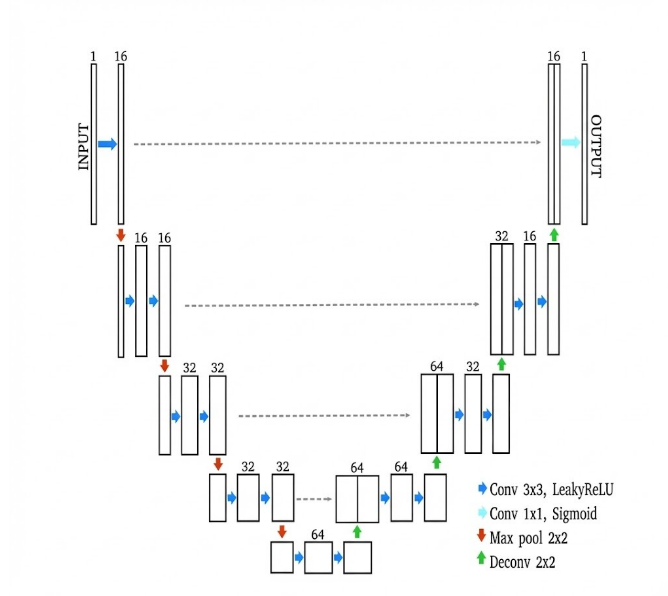


Fig. 1: Lightweight U-Net architecture.

4) Composite Noise Model: The final noisy image used for training combines all components:

$$I_{\text{noisy}} = \frac{N_{\text{shot}} + N_{\text{ron}} + N_{\text{dc}}}{t \cdot \alpha}, \tag{5}$$

This formulation ensures that synthesized degradations reflect realistic CCD imaging pipelines and improves the generalization of downstream models. In real astronomical imaging, exposure times typically vary; however, a fixed exposure time of $t = 2.0$ was used to clearly isolate and study the behaviour of noise-specific denoising models without introducing additional variability.

C. U-Net Architecture

Each denoiser in the framework is implemented as a lightweight U-Net that employs a symmetric encoder–decoder architecture with skip connections, shown in Fig. 1. The encoder consists of three convolutional blocks that progressively reduce spatial resolution while increasing feature dimensionality to enable hierarchical representation extraction. A bottleneck module captures the latent features before the decoder mirrors this process using transposed convolutions to restore spatial resolution.

These skip connections directly connect the corresponding encoder and decoder layers, ensuring that high-frequency spatial details suppressed during downsampling get reintroduced during reconstruction. This mechanism is particularly crucial in astronomical imaging, where faint structures like nebular filaments and stellar boundaries need to be preserved. Finally, a single-channel normalized output is obtained through a final 1×1 convolution that completes the restoration pathway.

D. Noise-Specific Denoising Models

We trained four independent variants of the U-Net to specialize in suppressing individual components of CCD noise.

Algorithm 1: Physics-Guided Astronomical Image Denoising Pipeline

Input : Clean Hubble image set D, noise simulator $\mathcal{N}$
Output: Denoised output image I_{fused}

Step 1: Preprocessing
Convert each image to grayscale; resize with long edge $= 512$ px; extract 128×128 patches.

Step 2: Train Noise-Specific Denoisers
foreach *noise type* $\tau \in \{shot, readout, dark, composite\}$ **do**
  Generate noisy patches $I_\tau = \mathcal{N}_\tau(I)$;
  Train U-Net model $\mathcal{M}_\tau$ using L2 loss with (I_τ, I) pairs;

Step 3: Train Fusion Network
Concatenate outputs $[I_{SN}, I_{RON}, I_{DN}, I_{ASTRO}]$; learn 1×1 convolution weights W to map concatenated predictions to I_{clean}.

Step 4: Inference
Given a noisy input I_{noisy}, compute $O_\tau = \mathcal{M}_\tau(I_{\text{noisy}})$;
Concatenate $O = [O_{SN}, O_{RON}, O_{DN}, O_{ASTRO}]$;
Compute fused restoration: $I_{\text{fused}} = W * O$;
return I_{fused}

The model `UNet_SN` targets photon shot noise, `UNet_RON` mitigates electronic read-out noise, and `UNet_DN` suppresses thermally induced dark-current noise. We also train a variant, `UNet_ASTRO`, on composite noise to generalize to mixed-noise conditions. This modular approach to training ensures each network develops expertise aligned with a particular astrophysical degradation process.

E. Training Strategy

Training was done in a supervised manner from paired clean and synthetically corrupted Hubble patches. Patches of size 128×128 are sampled randomly from the clean training set; noise is dynamically added by the physics-based simulator described in Section III-B. Hence every epoch, the models see different noise realizations and different spatial variations to prevent overfitting on any particular noise distribution.

All networks were optimized using the Adam optimizer with fixed learning rate and default momentum settings. The L_2 reconstruction loss,

$$L = \|I_{\text{clean}} - I_{\text{pred}}\|_2^2, \tag{6}$$

was employed to enforce pixel-level consistency between predictions and their corresponding clean references. Mixed-precision arithmetic was employed to reduce computational overhead and further accelerate training on GPU hardware.

Finally, the trained noise-specific U-Net models were frozen and utilized as specialist feature generators. The PSNR was used to evaluate the reconstruction fidelity numerically, while the SSIM is used to assess the perceptual and structural coherence. Subsequently, a lightweight 1×1 convolutional fusion layer was trained using the same objective as in (6) to combine the predictions from all denoisers into a single output, utilizing complementary strengths learned from various astrophysical noise distributions.

F. Learnable Fusion Module

To synthesize strengths among the four specialized denoisers, a learnable 1×1 convolutional fusion layer was introduced. During inference, outputs from all U-Net models are channel-wise concatenated and passed through a fusion layer that learns optimal weighting coefficients:

$$I_{\text{fused}} = W * [I_{\text{SN}}, I_{\text{RON}}, I_{\text{DN}}, I_{\text{ASTRO}}] \tag{7}$$

The resulting fused reconstruction consistently outperforms individual models by leveraging complementary noise suppression capabilities.

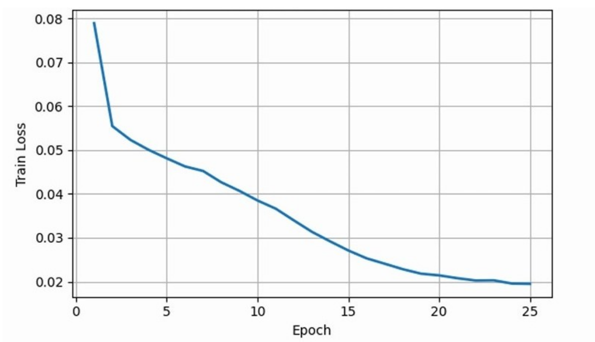


Fig. 2: Training loss convergence of the 1×1 fusion layer.

The training behavior of the 1×1 fusion module, shown in Fig. 2, reflects progressive refinement, with a steadily declining loss as the model learns to optimally weigh the outputs of noise-specific expert networks.

G. Overall Processing Pipeline

At inference time, a noisy astronomical image is independently forwarded through the four trained U-Net models, each specialized for a different noise component. The resulting outputs are concatenated along the channel dimension to form a multi-representation feature stack. The combined representation is then processed by the learnable 1×1 convolutional fusion module to integrate the complementary predictions and yield the final denoised reconstruction. With the exploration of noise-specific expertise and a unified fusion mechanism, this framework enables robust restoration performance under various astrophysical degradation conditions.

IV. Results and Discussion

The proposed denoising framework was evaluated on the processed Hubble dataset using synthetically corrupted images generated with the physics-driven noise model. Quantitative performance was measured by peak signal-to-noise ratio and the structural similarity index where all the reported values correspond to mean validation scores extracted from the

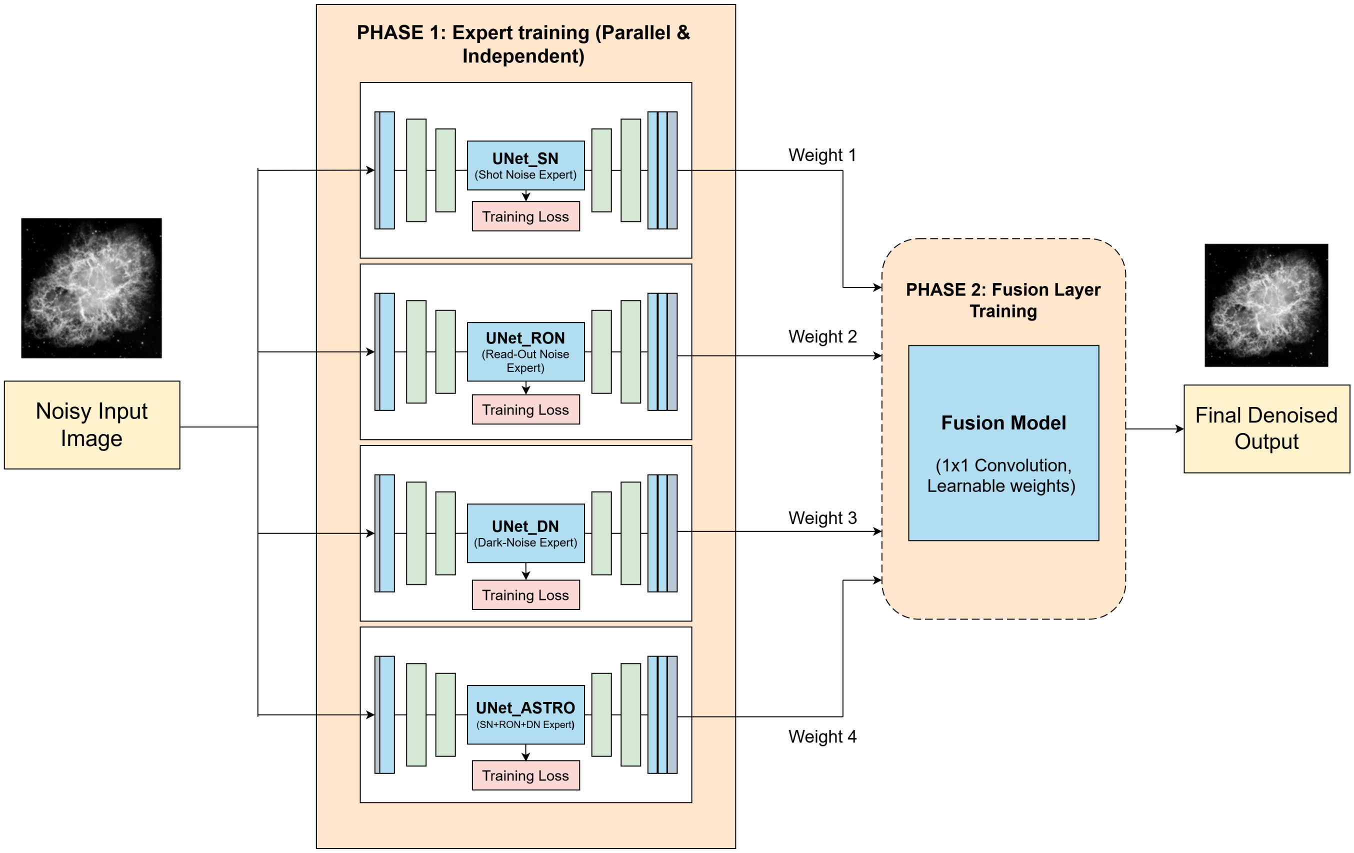
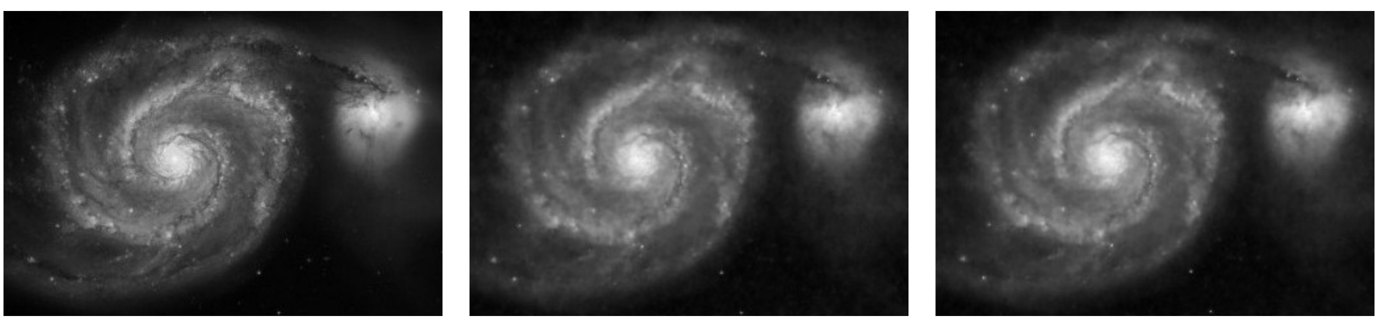


Fig. 3: Workflow of the proposed physics-guided multi-U-Net denoising framework, illustrating independent expert training for individual CCD noise components followed by frozen-weight feature fusion using a learnable 1×1 convolution.



(a) Clean reference image of a galaxy (Ground truth)

(b) `UNet_ASTRO` PSNR = 32.09 dB, SSIM = 0.809

(c) Fusion model PSNR = 32.34 dB, SSIM = 0.827

Fig. 4: Qualitative comparison of denoising results on a representative Hubble image.

recorded evaluation logs. A detailed summary of the obtained performance metrics could be found in Table I.

The results obtained by three independently trained noise-specific U-Net models, one for each of these isolated noise sources, are as follows: the shot-noise model, (UNet_SN), yielded a PSNR of 23.68 dB with an SSIM of 0.470 and the read-out noise model, (UNet_RON), resulted in 23.01 dB PSNR and an SSIM of 0.400. Nevertheless, among the single-noise experts, (UNet_DN) for dark-current noise had the best performance with 30.25 dB PSNR and an SSIM of 0.750. This trend is not surprising because thermal noise is more structured and predictable than either photon noise or electronic read-out noise.

To explore robustness under mixed-noise environments, a fourth network (UNet_ASTRO) was trained using images with all three astrophysical noise components simultaneously. This model achieved 31.16 dB PSNR and an SSIM of 0.820, setting a stronger baseline than any individual denoiser. The improvement confirms that jointly modeling Poisson–Gaussian noise interactions provides richer representational capacity

TABLE I: PSNR and SSIM performance of noise-specific U-Net models and the fusion approach.

Model	Description	PSNR (dB)	SSIM
UNet_SN	Trained on shot noise only	23.68	0.470
UNet_RON	Trained on read-out noise only	23.01	0.400
UNet_DN	Trained on dark-current noise only	30.25	0.750
UNet_ASTRO	Trained on all three noise components jointly	31.16	0.820
Fusion Model	Learnable 1×1 convolution combining outputs of all U-Nets	32.90	0.840

compared to treating each source in isolation.

The final stage employed a learnable 1×1 convolutional fusion layer to integrate the predictions of all four U-Net experts. This fused model achieved the highest reconstruction fidelity, achieving 32.90 dB PSNR and an SSIM of 0.840. This improvement demonstrates that the noise-specific U-Nets capture complementary signal characteristics, and that the fusion layer effectively leverages this diversity to yield a superior reconstruction.

Qualitative inspection, illustrated in Fig. 4 shows qualitative evidence supporting the quantitative results. In particular, the fused outputs always recover faint astrophysical structures such as nebular filaments and galactic arms that are still partially degraded in the predictions of individual models. The reconstructed images are less grainy, better spatially coherent, and clearer from a perceptual viewpoint for a wide variety of noise conditions. These findings clearly validate the benefits of physics-based modeling combined with a multi-expert deep learning architecture for astronomical image restoration.

V. Conclusion and Future Scope

The proposed framework considers a physics-guided astronomical image denoising approach, coupling domain-aware noise simulation with multiple U-Net architectures specialized for photon shot, read-out, and dark-current noise. Every model will be able to learn a particular degradation mechanism by using astrophysically realistic Poisson–Gaussian noise distributions based on CCD sensor behavior, while a composite U-Net generalizes across mixed-noise conditions. A lightweight 1×1 fusion module synergistically combines all the outputs of the denoisers, leading to a better PSNR and SSIM performance with enhanced perceptual fidelity. This combination of physically grounded noise modeling with multi-expert learning yields reconstructions of high quality that preserve faint celestial structures, reduce noise artifacts, and improve the interpretability of telescope imagery considerably.

Extending this to other astrophysical and sensor-induced distortions, such as cosmic ray interference, hot pixels, and atmospheric turbulence, will potentially allow for applications in both space- and ground-based observatories. Other future directions may include incorporating transformer or diffusion-based architectures to model broader contextual dependencies; integrating uncertainty quantification for scientific reliability; and enabling real-time denoising for instrument pipelines. Moreover, the denoised outputs coupled with downstream tasks like source extraction, photometric measurements, and morphological classification will not only validate the model's utility in scientific workflows but also establish a complete, end-to-end astronomical data enhancement ecosystem.

References

[1] J. R. Janesick, *Scientific Charge-Coupled Devices*. Bellingham, WA, USA: SPIE Press, 2001.

[2] R. J. Hanisch, R. L. White, R. A. Shaw, and K. R. Schmitt, *The Hubble Space Telescope Data Handbook*, vol. 1. Baltimore, MD, USA: STScI, 1997.

[3] M. Mäkitalo and A. Foi, "Optimal inversion of the generalized Anscombe transformation for Poisson–Gaussian noise," *IEEE Trans. Image Process.*, vol. 22, no. 1, pp. 91–103, 2013.

[4] A. Laine, "Noise reduction in astronomical images," *Astron. Astrophys.*, vol. 213, pp. 203–214, 1989.

[5] P. Perona and J. Malik, "Scale-space and edge detection using anisotropic diffusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 12, no. 7, pp. 629–639, 1990.

[6] M. Elad and M. Aharon, "Image denoising via sparse and redundant representations over learned dictionaries," *IEEE Trans. Image Process.*, vol. 15, no. 12, pp. 3736–3745, 2006.

[7] J. Mairal, F. Bach, J. Ponce, G. Sapiro, and A. Zisserman, "Non-local sparse models for image restoration," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2009, pp. 2272–2279.

[8] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, "Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising," *IEEE Trans. Image Process.*, vol. 26, no. 7, pp. 3142–3155, 2017.

[9] K. Zhang, W. Zuo, and L. Zhang, "FFDNet: Toward a fast and flexible solution for CNN-based image denoising," *IEEE Trans. Image Process.*, vol. 27, no. 9, pp. 4608–4622, 2018.

[10] K. Zhang, W. Zuo, S. Gu, and L. Zhang, "Learning deep CNN denoiser prior for image restoration," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 3929–3938.

[11] C. Chen, Q. Chen, J. Xu, and V. Koltun, "Learning to see in the dark," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 3291–3300.

[12] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *MICCAI*, 2015, pp. 234–241.

[13] N. Hu, J. Quan, and J. Gu, "U-Net for astronomical image denoising," *Mon. Not. R. Astron. Soc.*, vol. 493, no. 3, pp. 1–13, 2020.

[14] A. A. Elhakiem, M. E. Khaled, A. M. Abdelhafiz, and M. M. Fouad, "Astronomical image denoising based on convolutional neural network," *IEEE Access*, vol. 9, pp. 35670–35685, 2021.

[15] Y. Liu et al., "Multi-scale residual network for image restoration," *IEEE Trans. Image Process.*, vol. 28, no. 3, pp. 1127–1140, 2019.

[16] X. Li, Y. Zheng, and D. Liu, "Ensemble deep learning for image restoration," *IEEE Trans. Multimedia*, vol. 22, no. 3, pp. 704–716, 2020.

[17] R. Karim Sony, "Top 100 Hubble Telescope Images" [Dataset]. Kaggle. [Online]. Available: https://www.kaggle.com/datasets/redwankarimsony/top-100-hubble-telescope-images